

Asymptotic Notations

(P-1)

Function grow based on input:

OR 1 $1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < \dots < 2^n < 3^n < \dots < n^n$

Asymptotic Notations

(I) O big-oh $\rightarrow$ Upper bound

(II) Ω big-omega $\rightarrow$ lower bound

(III) Θ theta $\rightarrow$ Average bound

$\Rightarrow$ Time complexity of any algorithm follows the above order sequence **OR 1**

$\Rightarrow$ complexity is not among the **OR 1** then it may be multiple of them.

(I) Big-oh

The function $f(n) = O(g(n))$ iff $\exists$ +ve constants c and n_0 .

Such that

$$f(n) \leq c * g(n) \quad \forall \quad n \geq n_0$$

Example

$$f(n) = 2n + 3$$

$$2n + 3 \leq \text{--- (Something)} \leq 10n$$

$$2n + 3 \leq 10n$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ f(n) & c & g(n) \end{array}$$

$$n \geq 1$$

$$[n = 1, 2, \dots]$$

$$\therefore f(n) = O(n)$$

Again, $f(n) = 2n+3$

(P-2)

$$2n+3 \leq 7n$$

$$2n+3 \leq 2n+3n \leq 5n, \quad n \geq 1$$

Again

$$2n+3 \leq 2n^2+3n^2$$

$$2n+3 \leq 5n^2$$

$$\downarrow \downarrow$$
$$c \quad g(n)$$

$$n \geq 1$$

also true

Then, $f(n) = O(n^2)$

also true

QRI

$$1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < \dots < 2^n < 3^n < \dots < n^n$$

lower bound

upper bound

belongs to
Average
bound

Here

Eq 1

$f(n) = O(n)$ is true

$f(n) = O(n^2)$ is true

$f(n) = O(2^n)$ is true

But, I can not say

$$f(n) = O(\log n)$$

Because, $\log n$ is from lower bound

Therefore, we need to write the closest function
from Eq 1 that is

$$f(n) = O(n).$$

Omega notations

the function

$$f(n) = \Omega(g(n)) \text{ iff } \exists +ve \text{ constants } c \text{ and } n_0.$$

Such that

$$f(n) \geq c \cdot g(n) \quad \forall n \geq n_0.$$

Example

$$f(n) = 2n + 3$$

$$2n + 3 \geq 1 \times n \quad \forall n \geq 1$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ f(n) & c & g(n) \end{array}$$

Again

$$2n + 3 \geq 1 \times \log n$$

I can say $f(n) = \Omega(n)$ is true
 $\forall n \geq 1 \quad f(n) = \Omega(\log n)$ is true
But, we cannot write,

$$\times \quad f(n) = \Omega(n^2)$$

this is wrong -

we need to write the closest value, i.e.,

$$f(n) = \Omega(n).$$

Theta notation

(p-4)

the function $f(n) = \Theta(g(n))$ iff $\exists$ +ve const
 c_1, c_2 , and no.

Such that

$$c_1 * g(n) \leq f(n) \leq c_2 * g(n)$$

Example

$$f(n) = 2n + 3$$

$$1 * n \leq 2n + 3 \leq 5 * n$$

$$\begin{array}{ccc} c_1 * g(n) & \downarrow & \downarrow g(n) \\ & f(n) & c_2 \end{array}$$

I can say $f(n) = \Theta(n)$

* This is the average bound
of the function.

But, I can not write $f(n) = \Theta(n^4)$

We can use only n here.

We can use any notation for worst-case and
best case

* It is wrong to assign O -big-oh for
worst case and Ω -big-oh for best
case.